Semi-Supervised Remote-Sensing Image Scene Classification Using Representation Consistency Siamese Network

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***Abstract*— Deep learning has achieved excellent performance in remote-sensing image scene classification, since a large number of datasets with annotations can be applied for training. However, in actual applications, there is just a few annotated samples and a large number of unannotated samples in remote-sensing images, which leads to overfitting of the deep model and affects the performance of scene classification. In order to address these problems, a semi-supervised representation consistency Siamese network (SS-RCSN) is proposed for remote-sensing image scene classification. First, considering intraclass diversity and interclass similarity of remote-sensing images, Involution-generative adversarial network (GAN) is utilized to extract the discriminative features from remote-sensing images via unsupervised learning. Then, Siamese network with a representation consistency loss is proposed for semi-supervised classification, which aims to reduce the differences of labeled and unlabeled data. Experimental results on UC Merced dataset, RESICS-45 dataset, aerial image dataset (AID), and RS dataset demonstrate that our method yields superior classification performance compared with other semi-supervised learning (SSL) methods.**

***Index Terms*— Involution-generative adversarial network (GAN), remote-sensing image, scene classification, semi-supervised learning (SSL), Siamese network.**

I. Introduction

REMOTE-sensing images are acquired for earth observation, which can effectively measure and observe the detailed information of the earth surface [1]. With the development of remote-sensing technology, the spatial resolution of remote-sensing images is increasing rapidly [2]–[4]. High-resolution remote-sensing images not only have rich spectral, shape, and texture information, but also contain scene semantic information [5]–[8]. Scene classification of remote-sensing images is to effectively understand the scene and classify to a semantic category, which has been widely applied in urban planning, natural disaster detection, environmental monitoring, vegetation mapping, and so on [9].

In recent years, deep learning has made remarkable progress in computer vision, which has also gradually applied for remote-sensing image scene classification [10]–[12]. Convolutional neural network (CNN) has been introduced to remote-sensing image analysis around 2015 [13]. Compared with traditional methods, CNN has the advantage of end-to-end feature learning, as well as extracting effective features for image classification [14], [15]. Compared with low-level feature descriptors, CNN has achieved significant success in many applications. Chaib *et al.* [16] fused the deep features extracted by Visual Geometry Group Net (VGGNet) to improve the scene classification performance. He *et al.* [17] proposed a novel multilayer stacked covariance pooling (MSCP) algorithm for scene classification of remote-sensing images, which can automatically extract multi-layer feature maps from pre-trained CNN. Lu *et al.* [18] presented a feature aggregation convolutional neural network (FACNN) for scene classification, which learns scene representation by exploring semantic label information. Jun *et al.* [19] provided a skip linked covariance network, which embeds the skip linked and differential pooling modules into the traditional CNN model, combines the multi-resolution feature map, and makes full use of the second-order information to obtain representative feature. In order to solve the problem of scale variation of objects in remote-sensing images, Liu *et al.* [20] introduced multi-scale CNN for scene classification. Xie *et al.* [21] designed an adaptive CNN for scene classification, which can accept images of any size as the input without resizing.

Current scene classification algorithms are mainly supervised learning methods with large number of labeled data [20], [22]. These algorithms are time-consuming and requiring a great number of manual annotations. Data annotation is quite tough for remote-sensing images [22]. With insufficient training samples, it may result in overfitting for a deeper neural network, where the trained model yields poor performance on the test data with similar distribution. Moreover, when the deep network is insufficiently trained with limited labeled data, feature expression adaptability of the deep model is not strong, and thus samples of the same category will have inconsistent feature representations.

To overcome above issues of insufficient annotated data, semi-supervised classification has been studied for remote-sensing images. Wei *et al.* [23] proposed a semi-supervised generation framework (SSGF), which combines deep learning features, self-labeling technology, and discriminant evaluation method for scene classification and dataset updating. It can effectively learn from unlabeled samples and improve the classification accuracy. Yan *et al.* [24] introduced semi-supervised

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learning (SSL) into generative adversarial networks (GANs), which can utilize the discriminator to learn effective features from both labeled and unlabeled data. In addition, a hybrid data enhancement method is introduced to augment the data for SSL. Dai *et al.* [25] proposed a novel semi-supervised method, which uses ResNet to extract the initial image features, adopts integrated learning strategy to establish effective representation, and finally applies supervised learning for scene classification. Yao *et al.* [26] proposed a self-stepping deep learning method with local manifold constraints, which can select easy samples from the generated pseudo samples iteratively. Zhang *et al.* [27] developed a cooperative framework with supervised and unsupervised branches for scene classification. In addition, a novel semi-supervised center loss is proposed to learn discriminative features from remote sensing images.

In this article, we propose a semi-supervised representation consistency Siamese network (SS-RCSN) for remote-sensing image scene classification. The proposed model learns an adaptive GAN by unsupervised learning and constructs the Siamese network with representation consistency regularization loss for scene classification. The main contributions of this article are summarized as follows.

1) A representation consistency Siamese network (SS-RCSN) is proposed for remote-sensing image scene classification, which can solve the problem of overfitting with limited training samples in SSL.
2) Involution-GAN model is developed for unsupervised feature learning, which can extract discriminative features and reduce the intraclass diversity as well as interclass similarity.
3) A representation consistency loss function is proposed to measure the similarity between labeled data and unlabeled data embedded in high-dimensional space. It can reduce the differences of features between labeled and unlabeled data, which contributes to the scene classification performance.

The rest of this article is organized as follows. Section II presents the related work. The proposed SS-RCSN for remote-sensing image scene classification is described in Section III. In Section IV, performance of the proposed method is analyzed in the experiments. Conclusions are finally summarized in Section V.

II. Related Work

In the deep learning framework, SSL and unsupervised learning have achieved a great performance for various applications. In this section, we will briefly describe existing methods.

A. Unsupervised Learning

Many researchers have studied unsupervised feature learning methods for remote-sensing scene classification [22], [28]. Specifically, many scene classification methods adopt the unsupervised autoencoders-based methods and GAN-based methods [22], which have been widely used in the field of computer vision. Autoencoders and GAN can automatically learn valuable representations from unlabeled data. Cheng *et al.* [29] utilized single hidden layer neural network and autoencoders to train more effective sparse representation. Han *et al.* [30] proposed a convolution sparse autoencoder to overcome the inadequate representation. Cheng *et al.* [31] proposed an autoencoder to learn mid-level visual features, which can improve the accuracy of scene classification. Du *et al.* [32] proposed a stacked convolutional denoising autoencoder, which realizes excellent classification performance. Yao *et al.* [33] integrated paired constraints into stacked sparse autoencoders to obtain more discriminative features for scene classification and semantic annotation tasks.

In addition, some researchers apply GAN to scene classification. Yu *et al.* [34] proposed a multiple-layer feature-matching GAN for scene classification. Long *et al.* [35] proposed an unsupervised attention-GAN for remote-sensing scene classification, which enhances the feature representation ability of discriminator and improves the loss function of the generator and the discriminator.

Although unsupervised learning can make the model easier for feature extraction, the current feature extraction methods may suffer from several problems. Due to the complexity of ground object and spatial distribution in high-resolution remote-sensing images, the current methods cannot effectively describe the visual information, resulting in poor classification performance. To improve the accuracy of scene classification, it is necessary to obtain discriminative features [22], [36].

B. Semi-Supervised Learning (SSL)

SSL has been successfully applied in many fields over the past decade [37]. When just a few labeled data can be available, SSL provides a way to use unlabeled data to improve the performance [38]–[41].

There are many methods for SSL, including consistency regularization, entropy minimization, and so on. Consistency regularization applies data augmentation by leveraging the idea that a classifier should output the same class distribution even after augmenting [42], [43]. Many SSL approaches rely on consistency regularization to force the model output to remain constant when the input is disturbed [37]. Another way for SSL is entropy minimization, which requires that the classifier outputs low-entropy predictions of unlabeled data [44]. Entropy minimization can be achieved via encouraging the low-entropy distribution of the unlabeled data. For example, Lee *et al.* [45] proposed the idea of pseudo-label, which uses predicted labels of unlabeled data and minimizes the entropy of the distribution of training data. In addition, some works have combined different semi-supervised methods [38]. For example, Berthelot *et al.* [37] proposed the MixMatch algorithm, which combines entropy minimization and consistency regularization as an unified loss function to generate "guess label" for each unlabeled example. Sohn *et al.* [46] proposed the FixMatch algorithm, which uses the model's prediction of weakly and strongly enhanced unlabeled images to generate pseudo-labels.

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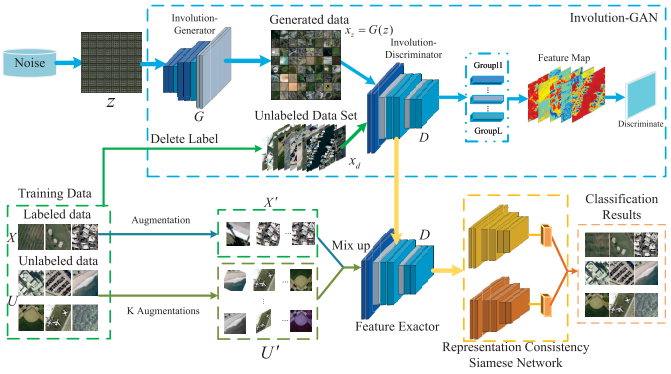


Fig. 1. Overall semi-supervised framework of the proposed model.

Although these methods achieve excellent performance, there are several problems of SSL for remote-sensing image scene classification. First, most of these methods use a single network, which has limitations in measuring data structure. Second, there are differences in features extraction of labeled data and unlabeled data. Finally, traditional loss functions cannot effectively cope with the problem of overfitting with limited samples.

III. Methodology

A. Overview

To solve the problem of overfitting with insufficient labeled samples, we propose an SS-RCSN for remote-sensing image scene classification. It consists of two parts: unsupervised Involution-GAN and Siamese network with a representation consistency loss function, which is shown in Fig. 1.

Assume the remote-sensing data as $V = \{v_1, \ldots, v_e, \ldots, v_s\}$, which can be divided into labeled data $X = \{x_1, \ldots, x_i, \ldots, x_n\}$ and unlabeled data $U = \{u_1, \ldots, u_j, \ldots, u_m\}$. Here, v_i represents the ith sample, $1 \le e \le s$, $1 \le i \le n$, $1 \le j \le m$, and $s = m + n$. The labels of the annotated data are $p = \{p_1, \ldots, p_i, \ldots, p_n\}$.

Considering intraclass diversity and interclass similarity in remote-sensing images, an unsupervised Involution-GAN is proposed to extract more discriminative features. Then, Siamese network with a representation consistency loss function is proposed, which can reduce the differences of features between labeled and unlabeled data. Finally, annotations of the dataset are updated to achieve semi-supervised classification of remote sensing images.

B. Involution-Generative Adversarial Network (GAN)

The major issues of remote-sensing images are intraclass diversity and interclass similarity, which seriously affect the performance of scene classification. Therefore, it is necessary to learn more discriminative feature representation. Inspired by this, we design a novel feature extraction that combines unsupervised learning and attention mechanism to extract more discriminative features. The Involution is used in our framework to construct a new architecture of deep network [38], which is divided into Involution-generator and Involution-discriminator.

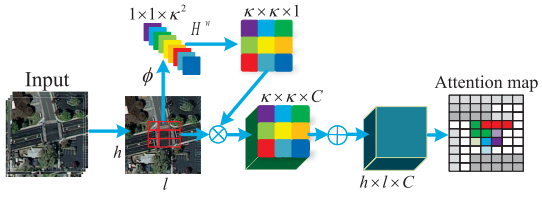


Fig. 2. Feature extraction process of the Involution layer.

The Involution structure has the characteristics of spatial particularity and channel invariance. Its feature extraction process is shown in Fig. 2. The shape of Involution kernel depends on the shape of input feature, and a large number of dynamic Involution kernels are used to collect rich receptive field information [38]. The output feature map of the remote-sensing image is obtained via performing the multiplication and addition operation, which is defined as [38]

$$Y_t^I = \sum H^w_{(\kappa,t,g,C_{\text{num}})} X_t^I \tag{1}$$

$$H^w = \phi(X^I) = \nu_1 \sigma(\nu_0 X^I) \tag{2}$$

where the input image size is $h \times l$, X_t^I stands for the input feature, H^w is the Involution kernel, g is the number of groups of the shared channel, κ represents the adjacent area, t represents feature channel, C_{num} is the number of feature channel, $\phi(\cdot)$ denotes the convolution kernel function, and σ stands for the activation function. ν_1 and ν_0 represent the weight of the linear transformation and Y_t^I is the output feature of remote-sensing image.

In the Involution-GAN, we introduce an Involution layer into the basic deep convolutional GAN (DCGAN), which is shown in Fig. 1. Specifically, an additional Involution layer is inserted to the layer of the generator, which constructs the Involution-generator G. At the same time, an additional Involution layer is inserted to the layer of the discriminator, which constructs Involution-discriminator D. Then, all the data V are utilized to train Involution-GAN in an unsupervised learning way. The output of the discriminator D is used to distinguish between the real data and generated data, and the generator G is used to generate remote-sensing images. After the training, the Involution-discriminator D can be used to extract features from the remote-sensing images.

During the training process, all the samples $V = \{v_1, \ldots, v_e, \ldots, v_s\}$ without labels are utilized for unsupervised learning. The noise z which follows a prior noise distribution $p_z(z)$ is fed into the generator G, which generates sample $x_z = G(z; \theta_g)$ with parameters θ_g. Here, Gaussian distribution is utilized as the noise distribution. The discriminator D aims to distinguish samples created by the generator and gives a probability $D(x; \theta_d)$, indicating the probability if the sample is the real data. The objectives to train the discriminator D are as follows: 1) maximizing $D(x_d)$ of each remote-sensing image x_d from the real training samples V and 2) minimizing $D(x_z)$ for each image in the false samples obtained from the generator. To sum up, training Involution discriminators requires maximizing the objective function

$$l_D = E_{x_d \sim p(x_d)} \log D(x_d) + E_{z \sim p_z(z)}[\log(1 - D(G(z)))]. \tag{3}$$

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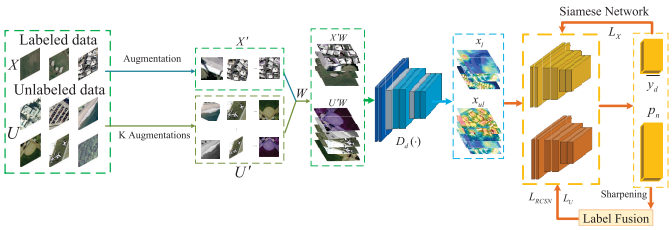


Fig. 3. Siamese network with a representation consistency loss function.

When the generator is trained, the weight of the discriminator can be fixed. The goal of the training generator $G(z)$ is to generate samples similar to the original image. Therefore, the generator wants to maximize the probability estimate of $D(G(z))$ and then minimize $1 - D(G(z))$

$$l_g = E_{z\sim p_z(z)}[\log(1 - D(G(z)))]. \tag{4}$$

The total loss function to train Involution-GAN is defined as follows:

$$\min_G \max_D V(D, G) = E_{x_d\sim p(x_d)} \log D(x_d) + E_{z\sim p_z(z)}[\log(1 - D(G(z))]. \tag{5}$$

Involution-GAN is trained to generate features that close to the real image features. For scene classification, the discriminant model D is regarded as the feature extractor D_d. At the same time, the generator G can also provide additional training data so that the discriminator can learn better representations.

C. Semi-Supervised Siamese Network

In actual applications, there is just a few annotated samples utilized for remote-sensing image scene classification, which greatly limits the classification accuracy. In order to solve this problem, we construct Siamese network Net_1 and Net_2 for semi-supervised classification, which aims to measure the similarity of labeled and unlabeled data embedded in the high-dimensional space. At the same time, a representation consistency loss is proposed to eliminate the differences between labeled and unlabeled data.

Data augmentation is applied both on the labeled and unlabeled data, which can be shown in Fig. 3. During the training process of the Siamese network, the labeled data X are augmented as X', whose labels are marked as p. The unlabeled data U are augmented as U'. X' and U' are mixed and randomly arranged to obtain the dataset W. The final input of the Siamese network is divided into two parts: one is mixed with X' and W, and the other is mixed with U' and W. Labeled data and unlabeled data are mixed up between different classes, which aims to expand the training dataset. Mix up is able to enhance the robustness of prediction on the data other than the training set.

For labeled samples X' of remote-sensing data, it is assumed that p_{d1} and p_{d2} are combined predictions of Net_1 and Net_2, and the corresponding labels are denoted as p. The trained feature extractor D_d is adopted to obtain the discriminative feature $x_l = D_d(X')$. The result of forward propagation $\overline{y}_d$ is defined as follows:

$$\overline{y}_d = w \cdot p_{d1} + (1 - w) \cdot p_{d2} \tag{6}$$

where w is the hyperparameter.

For the unlabeled samples, the augmented data U' is input to obtain the depth feature $x_{ul} = D_d(U')$. Finally, the weighted average output is taken as the forward propagation result p_n

$$p_n = \sum P_{\text{Net}_1}(y_1; \theta_1) + \sum P_{\text{Net}_2}(y_2; \theta_2) \tag{7}$$

where P_{Net_1} and P_{Net_2} are the predictions of unlabeled data by the Siamese network, and θ is the network parameter.

The forecast results of unlabeled data in remote-sensing images are sharpened to obtain pseudo-labels q. Specifically, the sharpening operation is written as follows:

$$\text{Sharpen}(q, T)_j = \frac{\left(1/K \sum P(U; \theta)\right)^{\frac{1}{T}}}{\sum_{j=1} \left(1/K \sum P(U; \theta)\right)^{\frac{1}{T}}} \tag{8}$$

where T is the sharpening parameter, K denotes the number of augmentation, and $P(U; \theta)$ is the prediction probability for each category.

The pseudo-label q predicted by Siamese network is fused as follows:

$$q = \max(\lambda, 1 - \lambda) q_{\text{Net}_1} + (1 - \max(\lambda, 1 - \lambda)) q_{\text{Net}_2} \tag{9}$$

where q_{Net_1} is the pseudo-label sharpened by the network Net_1, q_{Net_2} is the pseudo-label sharpened by the network Net_2, $\lambda \sim \text{Beta}(\varepsilon, \varepsilon)$, and ε is set to obey the probability distribution according to the actual data.

D. Representation Consistency Loss Function

To enhance the performance of the SS-RCSN model, a novel loss function is proposed for network training, which aims to reduce the differences of features between the labeled and unlabeled data. Unlabeled samples also contain potential information of data structure, which are utilized to train the network with stronger generalization ability. Furthermore, the average differences of features between the labeled and unlabeled data are measured, and the regularization of each category is calculated. Thus, the representation consistency loss function can be defined as follows:

$$L_{\text{RCSN}} = \frac{1}{c} \sum_{F=1}^{c} \|\varsigma\|^2{}_F + \frac{1}{c} \sum_{F=1}^{c} \|\upsilon\|^2{}_F \tag{10}$$

$$\varsigma = \frac{1}{n} \sum_{i=1}^{n} \text{Net}_1(x_{i-l}) - \frac{1}{m} \sum_{j=1}^{m} \text{Net}_2(x_{j-ul}) \tag{11}$$

$$\upsilon = \frac{1}{n} \sum_{i=1}^{n} \text{Net}_2(x_{i-l}) - \frac{1}{m} \sum_{j=1}^{m} \text{Net}_1(x_{j-ul}) \tag{12}$$

where c is the number of categories, Net_1 and Net_2 denote the Siamese networks, x_l and x_{ul} are the depth features with respect to labeled and unlabeled data, and F denotes the category index. In the specific calculation process, it can be

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expressed as

$$\left\| \frac{1}{n}\sum_{i=1}^{n} \mathrm{Net}_1(x_{i-l}) - \frac{1}{m}\sum_{j=1}^{m} \mathrm{Net}_2(x_{j-ul}) \right\|^2 = \left\| \frac{1}{n^2}\sum_{i=1}^{n} \xi - \frac{2}{nm}\sum_{i=1}^{n} \psi + \frac{1}{m^2}\sum_{j=1}^{m} \zeta \right\|^2 \tag{13}$$

$$\xi = \sum_{i'=1}^{n} k\left(x_{i-l}, x'_{i-l}\right) \tag{14}$$

$$\psi = \sum_{j=1}^{m} k\left(x_{i-l}, x_{j-ul}\right) \tag{15}$$

$$\zeta = \sum_{j'=1}^{m} k\left(x_{j-ul}, x'_{j-ul}\right) \tag{16}$$

where k denotes the Gaussian kernel function.

The cross-entropy loss function is applied on the labeled data, and the proposed loss function is applied on the unlabeled data. For the augmented labeled data X', the loss function can be established as follows:

$$L_X = \frac{1}{|X'|} \sum_{x,p \in X'} H(p, \overline{y}_d(x)). \tag{17}$$

For the augmented unlabeled data U', the loss function is established as follows:

$$L_U = \frac{1}{c|U'|} \sum_{u,q \in U'} \|q - p_n(u)\|_2^2 \tag{18}$$

where $\left|X'\right|$ denotes the number of labeled samples in each batch, $\left|U'\right|$ denotes the number of unlabeled samples in each batch, $H(P, \overline{y}_d)$ is the cross entropy function, x and p stands for the augmented samples and labels, respectively, and u and q are the augmented unlabeled data and pseudo-labels, respectively.

Therefore, the overall loss function to optimize the proposed SS-RCSN model can be written as follows:

$$L = L_X + \lambda_U L_U + \lambda_R L_{\mathrm{RCSN}} \tag{19}$$

where λ_U and λ_R are hyperparameters.

Algorithm 1: SS-RCSN model

Input: Remote sensing images V, hyperparameters ω, λ_U, λ_R, β_U, the number of groups of the shared channel g, the number of feature channel C_{num}, batch of labeled examples X and their labels P, batch of unlabeled examples U, sharpening temperature T, number of augmentations K, noise examples $z = \left(z^1, z^2, \ldots, z^s\right)$ from data prior $P_{prior}(z)$

Output: The trained SS-RCSN model.

Input noise examples z into Involution-generator G ;
Obtain generated data $x_z = G(z; \theta_g)$ by Involution-generator G;
Input noise examples z and remote sample $x_d = \upsilon$ into Involution-discriminator D ;
Update the parameters of Involution-discriminator D to maximize l_D;
Update the parameters of Involution-generator parameters D to minimize $\min_G \max_D V(D, G)$;
Obtain enhanced data X' and U' by applying data augmentation to labeled data X and unlabeled data U ;
Combine and shuffle enhanced labeled data X' and unlabeled data U';
Obtain input data W through the mix up of enhanced data;
Obtain the features x_l and x_{ul} of input data by the trained Involution-discriminator D_d;
Obtain the forward propagation results P of the Siamese network Net_1 and Net_2 by (7);
Obtain the pseudo label q of unlabeled data by (8) and (9);
Optimize the proposed SS-RCSN model by (19);

IV. Experiments

A. Datasets

To verify the effectiveness of the proposed method, four public datasets for remote-sensing image scene classification are applied in the experiments, including traditional learning methods, unsupervised feature-based methods, and SSL methods. They are UC-Merced Land Use dataset (UCM), NWPU RESICS-45 dataset, aerial image dataset (AID) [22], and the merged RS dataset [12], [47], [48], which are shown in Figs. 4–7, respectively.

UC-Merced Land Use dataset was obtained by U.S. Geological Survey, which reflects the urban area with the pixel resolution of 1 ft. There are 21 categories of scene images with the pixel size of 256×256, and each category has 100 images.

AID data was released by the Huazhong University of Science and Technology, Wuhan, China, and Wuhan University, Wuhan, in 2017, which contains 30 categories of scene images with the pixel size of 600×600. There are about 200–400 samples in each category.

NWPU-RESISC45 dataset was released by Northwestern Polytechnical University, Xi'an, China, which contains 31 500 images with the pixel size of 256×256. There are 45 categories and 700 images in each category.

The merged RS image scene dataset is a combined dataset by integrating five public datasets including UCM, AID, NWPU-RESISC45, RSI-CB256, and PatternNet, which is composed of 70 scene categories, and each category contains 800 image scenes with a size of 256×256 [12], [47], [48].

B. Parameter Setup

In the experiments, the proposed model is compared with the traditional methods, the supervised methods, the unsupervised methods, and the semi-supervised methods. Each dataset is split into 60% training set, 20% validation set, and 20% test set.

In the training set, 10% data are set as labeled data and the remaining data are set as unlabeled data. In the proposed

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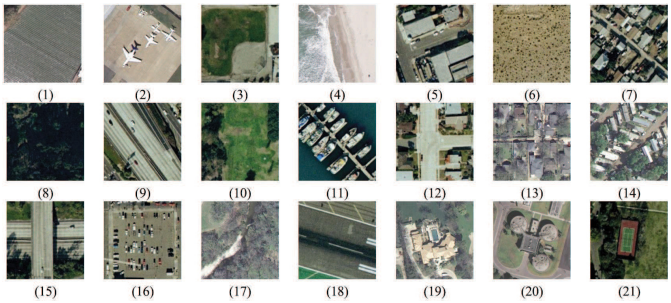


Fig. 4. UCM Land Use dataset, where (1)–(21) denote the category of agriculture, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis courts, respectively.

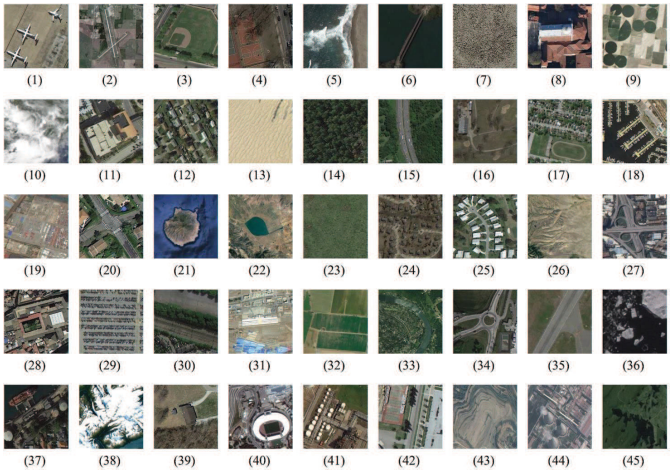


Fig. 5. RESICS-45 dataset, where (1)–(45) denote the category of airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland, respectively.

model, hyperparameters w, λ_U, λ_R, and β_U are all set to 0.5, the learning rate is set to 0.0005, and the batch size is set to 32. The group channel G is set to 4. The reduction ratio r is set to 4. The Adam optimizer is employed to train the proposed deep model. All the models are trained for 120 epochs.

1) *Involution-GAN:* The framework is shown in Fig. 8. The generator and the discriminator in Involution-GAN are both composed of six layers, where an Involution layer is both added to the generator and the discriminator.
2) *Data augmentation:* During the training process, data augmentation is adopted in the SS-RCSN model. Considering the scale difference of the labeled data and unlabeled data, data augmentation methods of these two streams are different. For labeled data, randomly cropping 224×224 patches from the original images is applied, and followed by horizontal flipping. For unlabeled data, randomly cropping, random rotation, and color jitter are applied, and thus the parameter K is set to 3. The predicted results of these three methods are averaged as the pseudo labels.

In addition, all the experiments were conducted on the server equipped with Intel CPU i9-9960X, NVIDIA Geforce 2080Ti GPU, and 128-GB DDR4 memory. ResNet18 is applied as the classification backbone.

C. Comparisons on UCM Dataset

In order to analyze the effectiveness of Involution-GAN, the feature map from Involution-GAN is outputted, and then t-distributed random neighbor embedding algorithm (t-SNE) is applied to embed features into the 2-D space. The visualization results are shown in Fig. 9. It can be seen that features of same category become closer in 2-D space. Fig. 10 depicts some generated examples by Involution-GAN. The generated examples with more iterative loops are more realistic.

To visually illustrate the efficacy of Involution, feature maps of UCM dataset extracted using Involution-GAN and GAN without Involution layer in Figs. 11 and 12, respectively. According to the results, GAN without Involution layer performs higher response values on the non-core irrelevant areas of the scene. In contrast, the proposed Involution-GAN can accurately capture the major objects in each scene, which has a great contribution to scene classification. Therefore, according to intraclass diversity and interclass similarity in remote-sensing images, it is reasonable that our feature extraction model has a superior adaptive attention mechanism, which can learn more discriminative features. Moreover, since unsupervised learning is applied in our framework, it is able to achieve better performance when measuring the resemblance between samples, which also verifies the superior performance of Involution-GAN.

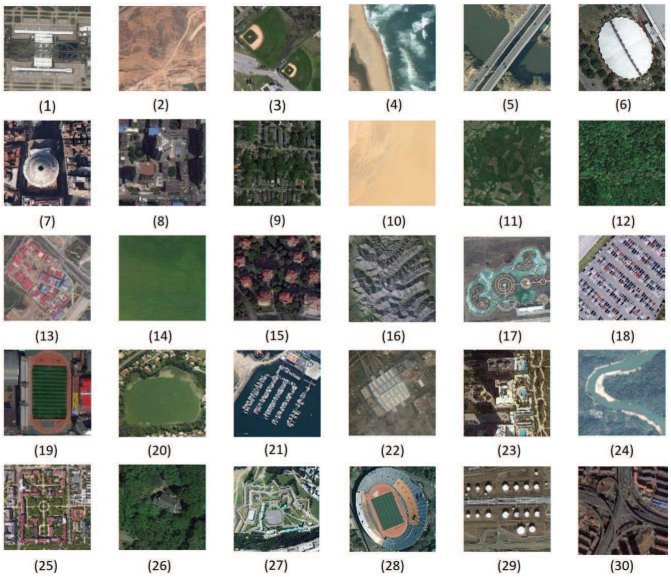


Fig. 6. AID dataset, where (1)–(30) denote the category of airport, bare land, baseball field, beach, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, port, railway station, resort, river, school, sparse residential, square, stadium, storage tank, and viaduct, respectively.

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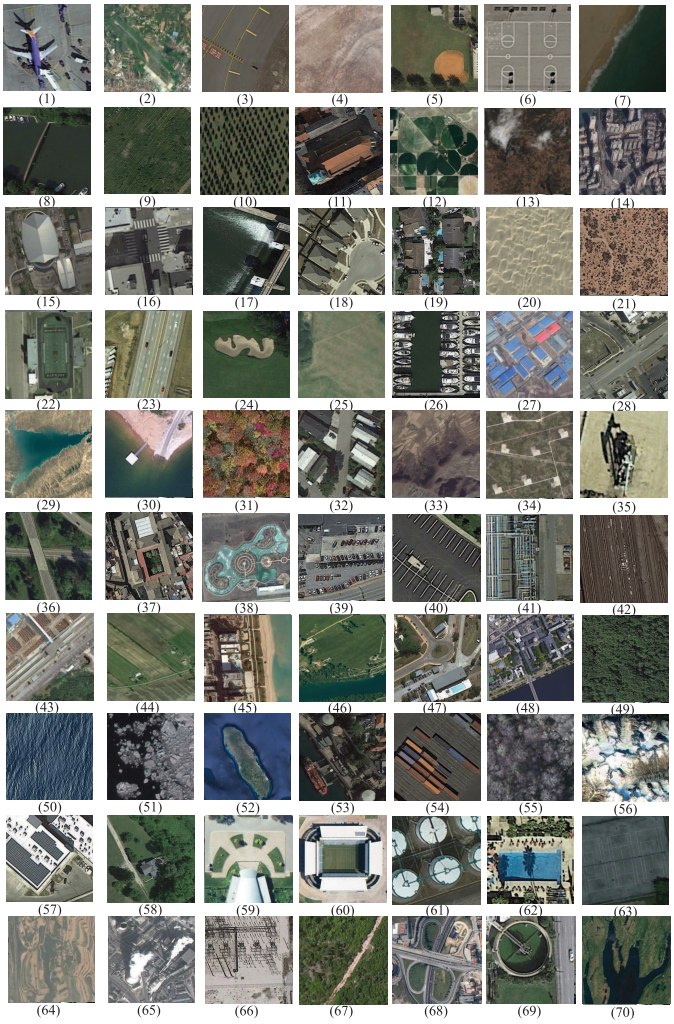


Fig. 7. Merged RS dataset, where (1)–(70) denote the category of airplane, airport, airport runway marking, bare land, baseball field, basketball court, beach, bridge, cemetery, Christmas tree farm, church, circular farmland, cloud, commercial area, convention center, crosswalk, dam, dead-end road, dense residential, desert, desert chaparral, football field, freeway, golf course, grass, harbor, industrial area, intersection, lake, lakeshore, mangrove, mobile home park, mountain, oil gas field, oil well, overpass, palace, park, parking lot, parking space marking, pipeline, railway, railway station, rectangular farmland, resort, river, roundabout, school, sclerophyllous forest, sea, sea ice, sea island, ship, shipping yard, shrubwood, snow mountain, solar panel, sparse residential, square, stadium, storage tank, swimming pool, tennis court, terrace, thermal power station, transformer station, tree lined path, viaduct hub, waste water plant, and wetland, respectively.

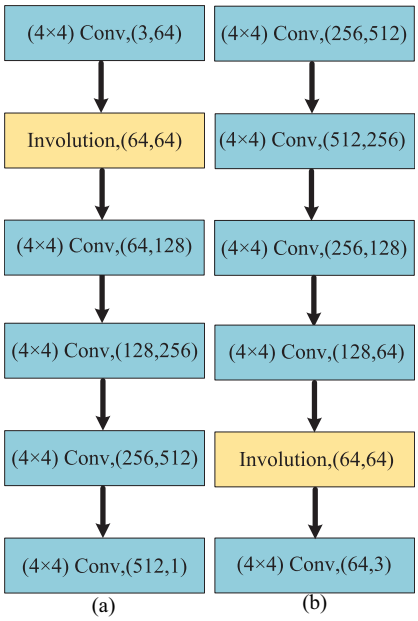


Fig. 8. Detail architecture of the generator and the discriminator. (a) Involution-discriminator. (b) Involution-generator.

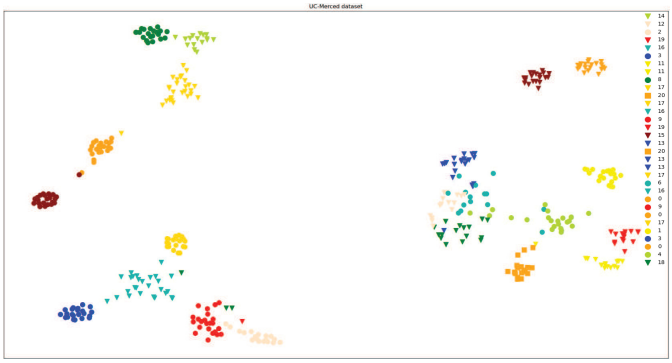


Fig. 9. Two-dimensional feature visualization of image global representations of the UC-Merced dataset. (0)–(20) stand for agricultural, airplane, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, baseball diamond, tennis court, beach, buildings, chaparral, dense residential, forest, freeway, and golf course, respectively.

Table I shows the accuracy comparisons on the UCM dataset. It can be seen that the proposed method yields higher accuracies than other semi-supervised approaches, which improves by 3.48%, 4.25%, 2.65%, and 1.88% compared with SSGA-E, Co-training, Mixmatch, and Fixmatch, respectively. Specifically, Mixmatch [37] achieve an accuracy of about 95.45%, and the accuracy of Fixmatch [46] is further increased by about 0.77%. Furthermore, our proposed method achieves the optimal results among all semi-supervised methods.

Compared with the supervised methods, accuracy of the proposed method is higher than the supervised methods with 10% and 50% training sets. When the supervised model using 80% training set, the proposed SS-RCSN model is slightly worse than the ARCNet-VGG16 model. Although our method just utilizes 10% labeled data, it can approximately achieve close accuracy to the supervised method using 80% labeled data. The reasons for such a great improvement is that our SS-RCSN can obtain discriminative features and reduce the differences of features between the labeled and unlabeled data. In addition, the proposed consistency regularization loss function can boost the ability of our deep network to learn from unlabeled data.

At the same time, to compare with the traditional methods, Dictionary Learning [49], Sparse Coding [50], and support vector machine (SVM) [51] are utilized for scene classification of remote-sensing images. From the compared results, the performance of our method is much higher than that of the traditional methods, because the traditional methods cannot learn valuable information from the unlabeled data.

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TABLE I
CLASSIFICATION RESULTS OF ALL METHODS ON THE UCM DATASET

	Method name	10% training set	50% training set	80% training set
Traditional methods	Sparse coding+SVM	45.64±1.02	52.61±1.59	58.56±1.29
	Dictionary learning+SVM	46.89±1.57	54.37±1.34	59.11±0.98
Supervised learning	GoogleNet [52]	83.64±2.29	-	-
	VGG-F [53]	85.61±1.06	-	-
	ResNet [54]	88.90±2.48	-	-
	ARCNet-VGG16 [55]	-	96.81±0.14	**99.12±0.40**
	GBN [56]	-	97.05±0.19	98.57±0.48
	ADFF [57]	-	96.05±0.56	97.53±0.63
Unsupervised learning	Attention-GAN [35]	-	89.06±0.50	97.69±0.69
	Self-training (ResNet) [23]		91.57±2.00	
Semi-Supervised learning	Co-training [58]	93.75±1.42		
	SSGA-E [23]	94.52±1.38		
	Fixmatch [46]	96.22±0.21		
	Mixmatch [37]	95.45±0.43		
	Our Method	**98.10±0.23**		

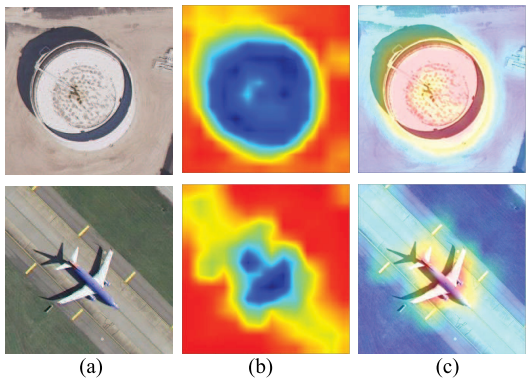
TABLE II
CLASSIFICATION RESULTS OF ALL METHODS ON THE RESICS-45 DATASET

	Method name	10% training set	60% training set
Traditional methods	Sparse coding+SVM	38.64±1.02	50.63±1.26
	Dictionary learning+SVM	39.53±0.96	50.91±1.78
Supervised learning	GoogleNet [52]	76.00±0.99	82.65±2.08
	VGG-F [53]	79.40±0.80	86.33±0.47
	VGG-19 [23]	78.49±1.14	85.24±0.11
	ResNet [54]	84.30±0.60	90.20±0.20
	BoCF [9]	82.65±0.31	-
	MSCP [17]	88.07±0.18	-
	CNN-CapsNet [59]	89.03±0.21	-
	SCCov [19]	89.30±0.35	-
Unsupervised learning	Attention-GAN [35]	72.21±0.21	-
	Self-training (VGG-S) [23]	81.46±0.68	
	Self-training (ResNet) [23]	85.82±1.30	
Semi-Supervised learning	Co-training [58]	87.25±0.95	
	SSGA-E [23]	88.60±0.31	
	Fixmatch [46]	90.45±0.43	
	Mixmatch [37]	89.22±0.29	
	Our Method	**92.03±0.16**	



(a) (b)

Fig. 10. Some generated examples from the UCM dataset by Involution-GAN. (a) Iterating 5000 times. (b) Iterating 50000 times.



(a) (b) (c)

Fig. 11. Feature map of UCM dataset extracted by Involution-GAN. (a) Original image. (b) Feature map. (c) Heat map.

D. Comparisons on RESICS-45 Dataset

Due to the size of the NWPU-RESISC45 dataset, we reduced the batch size ratio to 10% of the unlabeled dataset. Fig. 13 shows the results of embedding high-dimensional features into 2-D space by T-SNE. We can easily observe that in the distribution of 45 categories, the clustering of the same category is more centralized, and samples of different categories are dispersed, which will provide

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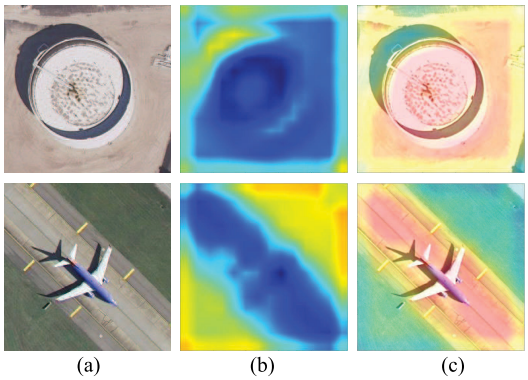


Fig. 12. Feature map of UCM dataset extracted by GAN without Involution layer. (a) Original image. (b) Feature map. (c) Heat map.

more effective feature expression for the classification of large-scale datasets. Fig. 14 shows remote-sensing scene data of RESICS-45 dataset generated by Involution-GAN.

Visual examples of Involution-GAN and GAN without Involution layer are shown in Figs. 15 and 16, respectively. According to the results, GAN without Involution layer cannot effectively capture the salient region of the objects. In this case, since the high intra-class diversity and inter-class similarity of the large-scale remote-sensing images, inaccurate positioning of objects will affect the classification performance. In contrast, the proposed Involution-GAN is able to capture the core region in each scene, which verifies the effectiveness of our model for feature extraction from large-scale remote-sensing images.

Table II shows the comparison results on the RESICS-45 data. From the table, we can see that the proposed method finally achieves 92.03% in the test set, which is the highest of semi-supervised methods. The proposed network produces higher accuracies than traditional methods, supervised learning methods, unsupervised learning methods, and SSL methods. It is indicated that the proposed SS-RCSN model is suitable for large-scale datasets, which improves the classification performance of scene classification for large-scale datasets.

E. Comparisons on AID Dataset

The parameter settings on the AID dataset are retained the same as on NWPU-RESISC45 dataset. Fig. 17 shows the result of embedding high-dimensional features into 2-D space. Similarly, in the data distribution of all categories, clustering of the same category is more centralized, and samples of different categories are dispersed mutually. Fig. 18 shows remote-sensing images generated by Involution-GAN. Figs. 19 and 20 are visual feature maps of scene "center" and "square" extracted by Involution-GAN and GAN without Involution layer, respectively. It is clearly seen that the proposed Involution-GAN can effectively locate the core location of saliency objects, which reflects the effectiveness of Involution-GAN in feature extraction.

Table III shows the overall accuracies on AID. Our proposed model achieves the best classification accuracy among all the results. The proposed semi-supervised framework and

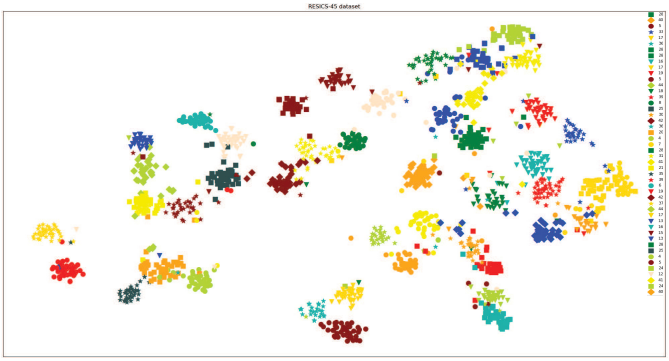


Fig. 13. Two-dimensional feature visualization of image global representations of the RESICS-45 dataset. (0)–(44) denote the category of airplane, airport, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, baseball diamond, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, basketball court, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snowberg, sparse residential, stadium, beach, storage tank, tennis court, terrace, thermal power station, wetland, bridge, chaparral, church, circular farmland, and cloud, respectively.

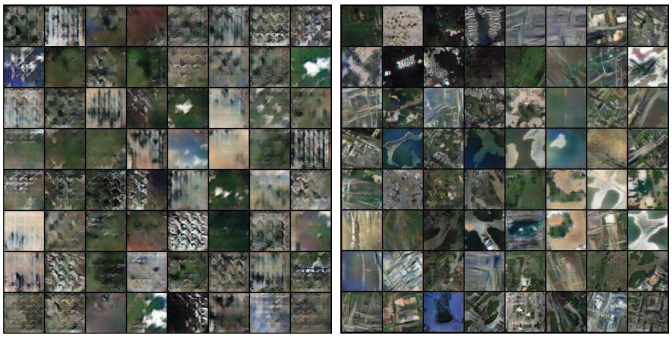


Fig. 14. Some generated example images from the RESICS-45 dataset by Involution-GAN. (a) Iterating 5000 times. (b) Iterating 50 000 times.

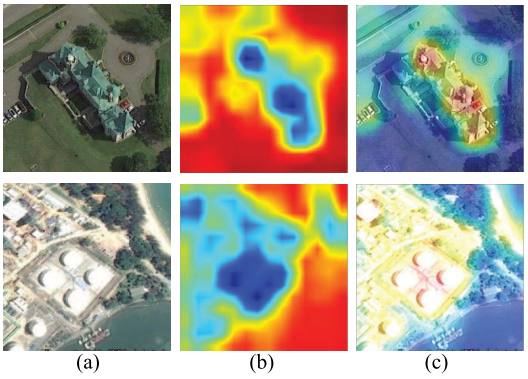


Fig. 15. Feature map of RESICS-45 dataset extracted by Involution-GAN. (a) Original image. (b) Feature map. (c) Heat map.

labeled consistency regularization loss function can enhance the performance of scene classification.

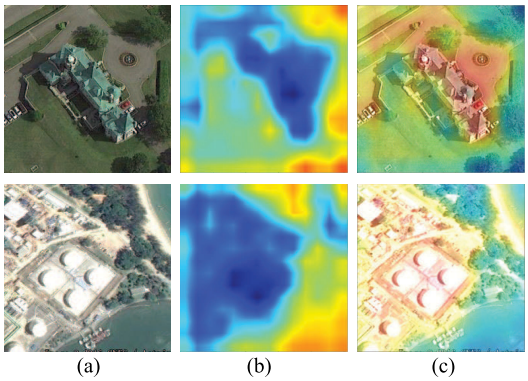
F. Comparisons on the Merged RS Image Scene Dataset

In order to prove the superiority of our method, we conducted experiments on a larger remote-sensing scene dataset.

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TABLE III
CLASSIFICATION RESULTS OF ALL METHODS ON THE AID DATASET

	Method name	10% training set	20% training set	60% training set
Traditional methods	Sparse coding+SVM	41.64±1.02	46.54±0.38	54.95±0.78
	Dictionary learning+SVM	42.31±0.87	47.23±0.56	55.93±1.57
Supervised learning	VGG-F [53]	82.25±0.95	-	89.28±0.87
	ResNet [54]	85.71±1.44	-	93.31±1.67
	GBN [56]	-	92.20±0.23	-
	DDRL-AM [60]	-	92.36±0.10	-
	MG-CAP [61]	-	93.34±0.18	-
Unsupervised learning	Attention-GAN [35]	-	78.95±0.23	-
	Self-training (ResNet) [23]		89.38±0.87	
Semi-Supervised learning	Co-training [58]	90.87±1.08		
	SSGA-E [23]	91.35±0.83		
	Fixmatch [46]	93.63±0.60		
	Mixmatch [37]	92.52±0.48		
	Our Method	**94.25±0.64**		



(a) (b) (c)

Fig. 16. Feature map of UCM dataset extracted by GAN without Involution layer. (a) Original image. (b) Feature map. (c) Heat map.



(a) (b)

Fig. 18. Some generated example images from the AID dataset by Involution-GAN. (a) Iterating 5000 times. (b) Iterating 50 000 times.

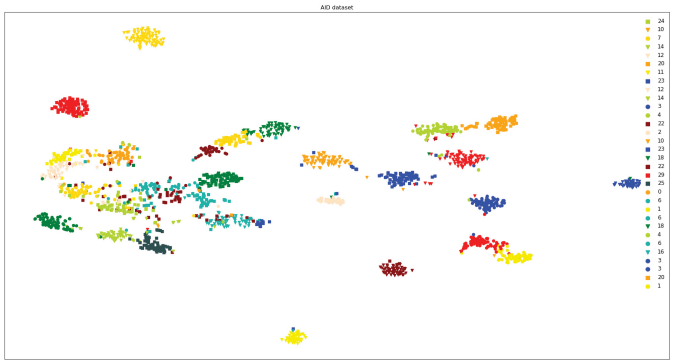
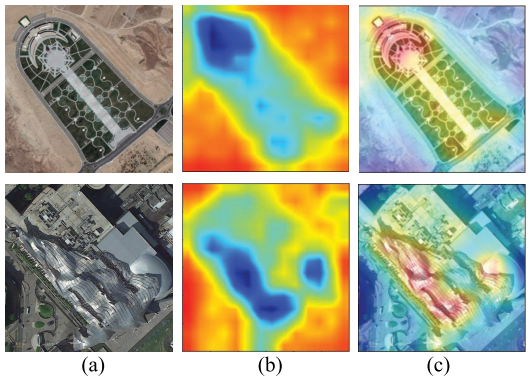


Fig. 17. Two-dimensional feature visualization of image global representations of the AID dataset. (0)–(29) denote airport, bare land, farmland, forest, industrial, meadow, medium residential, mountain, park, parking, playground, pond, baseball field, port, railway station, resort, river, school, sparse residential, square, stadium, storage tanks, viaduct, beach, bridge, center, church, commercial, dense residential, and desert, respectively.



(a) (b) (c)

Fig. 19. Feature map of AID dataset extracted by Involution-GAN. (a) Original image. (b) Feature map. (c) Heat map.

Table IV shows the overall accuracies on RS dataset using different proportion of unlabeled training data. Our proposed model achieves the best classification accuracy among all the results. In summary, it is proved that the proposed method is able to learn valuable information from unlabeled data for large-scale datasets and solve the problem of over-fitting with limited training samples.

G. Ablation Experiment

In order to better analyze our SSL model, an ablation study was conducted on the UCM dataset. We study the effect of removing components in the proposed SS-RCSN. The

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TABLE IV
CLASSIFICATION RESULTS OF ALL METHODS ON THE MERGED RS IMAGE SCENE DATASET

	Method name	10% training set	20% training set	60% training set
Semi-Supervised learning	Mixmatch [37]	90.89±0.32	91.01±0.67	91.35±0.21
	Fixmatch [46]	92.35±0.97	93.19±0.82	94.78±1.15
	Our Method	**92.25±0.38**	**93.25±0.53**	**95.25±0.47**

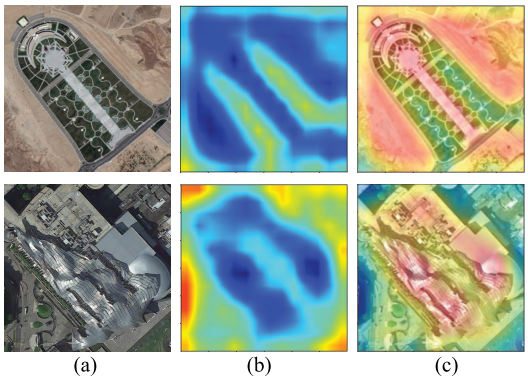


Fig. 20. Feature map of AID dataset extracted by GAN without the Involution layer. (a) Original image. (b) Feature map. (c) Heat map.

TABLE V
ABLATION EXPERIMENT RESULTS OF THE METHOD IN THIS ARTICLE. ✓MEANS ADD MODULE

Component	UCM dataset			
Involution-GAN	✓	×	✓	✓
Loss Fuction(SS-RCSN)	×	✓	✓	✓
Mix Up	✓	✓	×	✓
Accuracy(Acc)(%)	96.92	95.57	97.21	**98.10**

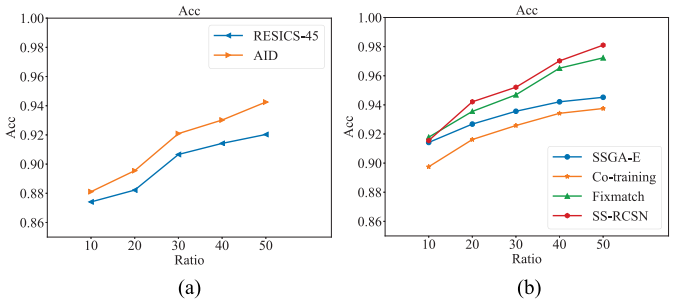


Fig. 21. Comparisons under different proportions of unlabeled data. (a) Comparisons of different methods. (b) Comparisons of different datasets.

experimental results are shown in Tables V–VII, respectively, where ✓means adding module.

1) Removing Involution-GAN: In the proposed feature extraction module, Involution-GAN is introduced to obtain more discriminative feature expression. On the other hand, the unsupervised training mode can eliminate the feature differences of labeled data and unlabeled data, which is helpful to improve the classification accuracy. For testing the role of Involution-GAN in the proposed model, Involution-GAN is removed, and the Siamese network is used for feature extraction and prediction. The test results are shown in Table V. It can be seen that the accuracy is greatly decreased when Involution-GAN is removed. The major reason is that the effective information cannot be extracted and the over-fitting tends to be intensified when removing Involution-GAN.

2) Removing the Representation Consistency Regularization Loss Function: In order to prove the efficiency of the proposed loss, we removed the representation consistency regularization loss function. The experimental results are shown in Table V. It can be seen that after removal of the proposed loss function, the accuracy results drop seriously. In contrast, using representation consistency regularization loss function model has been effectively improve performance. The reason is that the proposed loss function reduces the differences of labeled data and unlabeled data and enhances the ability to learn the unlabeled samples.

3) Removing Mix Up: The experimental results are shown in Table V. It is observed that adding mix up has slight effect to improve the accuracy on the UC-Merced dataset, which contributes to SSL.

4) Changing the Proportion of Unlabeled Data: We evaluate the impact of the proportion of unlabeled training data. The proposed method is compared with other semi-supervised methods, where classification accuracies with different proportion of unlabeled training data are reported in Tables VI and VII, respectively. Fig. 21 depicts the accuracy curve of comparisons under different proportions of unlabeled data. The accuracy of our method increased from 91.55% to 98.10%. It shows that all three algorithms can improve the ability of scene classification by learning from unlabeled data. Compared with other semi-supervised methods, our method has a small performance improvement when the unlabeled dataset is 10%. When the unlabeled set is 50%, our method has been greatly improved by eliminating the differences of labeled data and unlabeled data. Therefore, when the unlabeled dataset is 50%, the effectiveness of the proposed Involution-GAN and representation consistency Siamese network (RCSN) loss can be fully utilized. Results of Table VII are consistent with that of Table VI, which indicates the effectiveness of our proposed model for learning large-scale datasets with unlabeled data.

5) Calculation Time: In order to analyze computational efficiency of the proposed approach, the average training time (ATT) and test time are reported and compared with other SSL methods on the UCM dataset. The compared results of calculation time are listed in Table VIII. It can be seen that the proposed method took less ATT than Fixmatch and more ATT than Mixmatch. In addition, the proposed SS-RCSN processed a test image in 2.0 ms faster than the Fixmatch and 0.9 ms slower than the Mixmatch.

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TABLE VI
COMPARISONS UNDER DIFFERENT PROPORTIONS OF UNLABELED DATA (UCM)

Method name	10% training set	20% training set	30% training set	40% training set	50% training set
Co-training	89.75±1.27	91.62±0.93	92.58±0.78	93.42±1.32	93.75±1.42
SSGA-E	91.42±0.95	92.68±0.87	93.56±1.42	94.21±1.18	94.52±1.38
Fixmatch [46]	91.78±1.21	93.55±0.53	94.69±1.55	96.52±1.45	97.23±0.95
Our Method	91.55±0.14	94.21±0.32	95.21±0.54	97.02±0.13	98.10±0.23

TABLE VII
COMPARISONS UNDER DIFFERENT PROPORTIONS OF UNLABELED DATA (RESICS-45/AID)

Dataset	10% training set	20% training set	30% training set	40% training set	50% training set
RESICS-45	87.41±0.39	88.22±0.10	90.66±0.58	91.42±0.73	92.03±0.16
AID	88.11±0.46	89.55±0.34	92.09±0.67	93.02±0.13	94.25±0.64

TABLE VIII
CALCULATION TIME OF ALL METHODS ON THE UCM DATASET

	Method name	ATT (min)	Test time (ms)
Semi-Supervised learning	Fixmatch [46]	168.9	34.3
	Mixmatch [37]	141.5	31.4
	Our Method	142.3	32.3

V. CONCLUSION

This article proposes an SS-RCSN for remote-sensing image scene classification. The experimental results on UCM, RESICS-45, AID datasets, and the merged RS image scene dataset indicate that the proposed semi-supervised model achieves excellent results. It is verified that Involution-GAN can learn discriminative feature representations, so that the model can be more conveniently converted from semi-supervised classification task. Moreover, it is proved that the proposed representation consistency regularization loss can reduce the differences of labeled data and unlabeled data, which greatly promotes the improvement of classification accuracies. The combination of Involution-GAN and RCSN loss function further heighten the ability to learn from unlabeled data, which has a great contribution to improve the scene classification performance.

Since the mix-up method is used in our method, it may increase the complexity of the input. In our future work, we prefer to develop a stronger semi-supervised data combination method for scene classification.

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